The Four Brains of the Ambient System

The ambient system has four distinct roles:

1. **AmbientEmitter**: sound sources in the world
2. **AmbientProfile**: soundscape definitions (mix presets)
3. **AmbientZone:** triggers that activate profiles
4. **AudioManager:** the conductor

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1. AmbientEmitter — “I exist and I play a track”

Each AmbientEmitter:

- Has an AudioSource
- Has a **track name** (string or enum like "firesound")
- Registers itself with AudioManager on Start

Conceptually:

“Hi AudioManager, I am track = firesound, and here is my AudioSource.”

AudioManager stores this in something like:

trackName → list of emitters

So at runtime it knows:

- firesound → [emitter A, emitter B]
- birdssound → [emitter C]

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2. AmbientProfile — “This is what this place should sound like”

This is the soul of the system.

An AmbientProfile is a **data asset** that contains layers, like:

Layer:

Track = firesound

Volume = 1.0 ← target volume for this layer

Priority = 0 ← higher = kept when budget is exceeded

[Override Playback = false] ← enable to control emitter playback from this layer

RandomStartTime = true

PitchRange = (1.0, 1.0)

Another layer:

- Track = birdssound
- TargetVolume = 0.3
- FadeTime = 4.0

Conceptually, a profile says:

“In this area, here is the desired mix.”

When **Override Playback** is disabled (default), the emitter uses its own RandomStartTime and PitchRange settings. Enable it on a layer when you want a specific profile to control how an emitter starts (e.g., a tense combat profile might suppress random start times for a more urgent effect).

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3. AmbientZone — “When player enters, use this recipe”

The zone is just a trigger with a reference to an AmbientProfile.

OnTriggerEnter(player):

AudioManager.ApplyProfile(myProfile)

The zone does NOT care about emitters.
It just says: use this soundscape now.

Exit behavior is controlled by the Exit Action setting:

In **Stack** mode (the recommended default for zones), choose:

- AutoPop — automatically pops the zone's token when the player exits, using the exit fade duration. This is the default and the right choice for most zones.
- None — leaves the token on the stack indefinitely. You are responsible for calling AudioManager.PopAmbientToken(token) manually. Useful for scripted sequences. The Inspector warns you when this is selected to prevent accidental token leaks.
- StopFade / StopImmediate — pop with or without fade, same as AutoPop but with explicit naming for clarity.

In **Replace** mode, StopFade and StopImmediate call ClearAmbient; None and AutoPop leave the profile active (the player's ambient does not change on exit).

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4. AudioManager — “Apply the mix to the real world”

This is where the interesting logic happens.

When a profile is applied, AudioManager:

Step 1 — Build desiredVolumeByTrack

It reads the profile and builds a dictionary:

firesound → 1.0

birdssound → 0.3

wind → 0.0

Step 2 — Match profile tracks to emitters

For each registered emitter:

- Look at emitter.Track
- Look up desiredVolumeByTrack[Track]
- If found → fade AudioSource to that volume
- If NOT found → fade to 0 (or ignore)

This is how zones affect emitters anywhere in the scene.

Why emitters do NOT need to be inside zones

Because zones do NOT search the scene for emitters, the AudioManager already knows all emitters globally, zones just change the global mix.

- One fire sound can be heard across multiple zones
- A far-away waterfall can fade in gradually
- Etc

How multiple zones really work

Imagine player walking:

Forest Zone → Camp Zone → Cave Zone

Each zone applies a different profile:

ForestProfile:

- birds 1.0
- wind 0.4
- fire 0.0

CampProfile:

- birds 0.2
- wind 0.1
- fire 1.0

CaveProfile:

- birds 0.0
- wind 0.0
- drip 0.6

Each time:

- Same emitters
- Different recipe
- AudioManager fades volumes accordingly

Support

For further support, contact **snogdev@gmail.com**